2.1.5 Biological membranes

Membranes are fundamental to the cell theory. The structure of the plasma membrane allows cells to communicate with each other. Understanding this ability to communicate is important as scientists increasingly make use of membrane-bound receptors as sites for the action of medicinal drugs.

Understanding how different substances enter cells is also crucial to the development of mechanisms for the administration of drugs.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the roles of membranes within cells and at the surface of cells	To include the roles of membranes as: • partially permeable barriers between the cell and its environment, between organelles and the cytoplasm and within organelles • sites of chemical reactions • sites of cell communication (cell signalling).
(b)	the fluid mosaic model of membrane structure and the roles of its components	To include phospholipids, cholesterol, glycolipids, proteins and glycoproteins **AND** the role of membrane-bound receptors as sites where hormones and drugs can bind. *M0.2* HSW1
(c)	**(i)** factors affecting membrane structure and permeability **(ii)** practical investigations into factors affecting membrane structure and permeability	To include the effects of temperature and solvents. *M0.1, M0.2, M1.1, M1.2, M1.3, M1.6, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6* **PAG5, PAG8** HSW1, HSW2, HSW3, HSW4, HSW5, HSW6
(d)	**(i)** the movement of molecules across membranes **(ii)** practical investigations into the factors affecting diffusion rates in model cells	To include diffusion and facilitated diffusion as passive methods **AND** active transport, endocytosis and exocytosis as processes requiring adenosine triphosphate (ATP) as an immediate source of energy. *M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.11, M2.1, M3.1, M3.2, M3.3, M3.5, M3.6, M4.1* **PAG8** HSW1, HSW2, HSW3, HSW4, HSW5, HSW6

(e)	**(i)**	the movement of water across membranes by osmosis and the effects that solutions of different water potential can have on plant and animal cells	Osmosis to be explained in terms of a water potential gradient across a partially-permeable membrane.
	(ii)	practical investigations into the effects of solutions of different water potential on plant and animal cells.	*M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.10, M1.11, M2.1, M3.1, M3.2, M4.1* **PAG8** HSW1, HSW2, HSW3, HSW4, HSW5, HSW6